WEEK – 2

Minor Project (BTCS607N)-Weekly Progress Report

Project Title: Drone-Based Intelligent System for Apple Orchard Management in Himachal Pradesh.

Group ID: G9

Branch & Section: BTech CSE (AIML with Microsoft), AIDS -A

Date: 30-01-2026

1. Executive Summary

A. **Current Sprint:** Sprint 1 (Issue – 2)

B. **Sprint / Issue Goal:** To identify system actors, define functional requirements, and design UML Use Case Diagram along with detailed use case descriptions.

C. **Status:** On Track

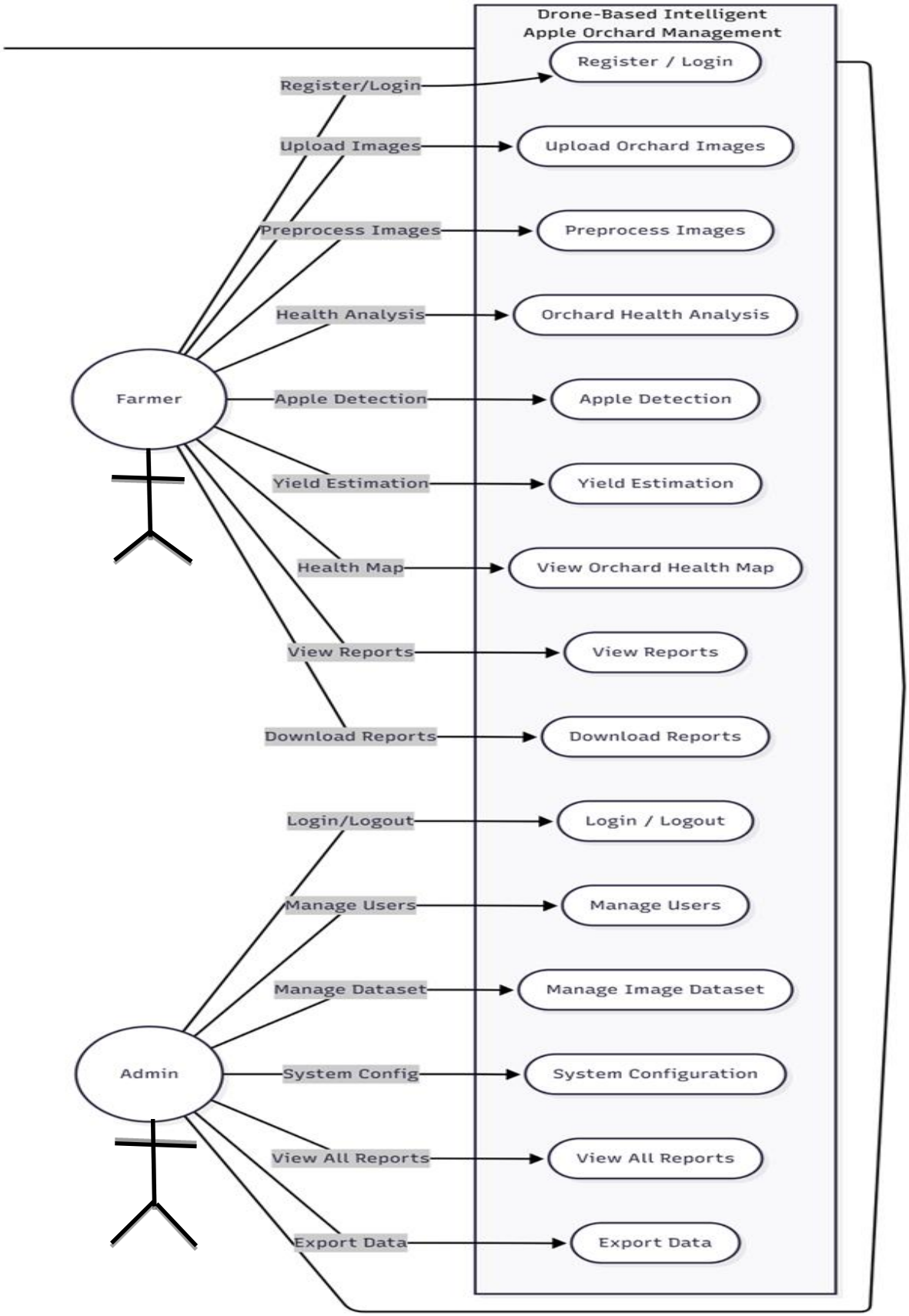
2. UML & OOAD Progress

A. **Diagrams Completed this week:** UML **Use Case Diagram** for the *Drone-Based Intelligent Apple Orchard Management System*

B. **Diagrams Revised:** Minor corrections made to improve clarity of the Use Case Diagram (Renamed some use cases, Fixed actor connections, Added / removed a use case)

C. **OO Concept Applied:**

1. Identified **actors** (Farmer, Admin)
2. Defined **use cases** for system functions
3. Linked actors with relevant use cases


Drone-Based Intelligent
Apple Orchard Management
Register / Login
Upload Orchard Images
Preprocess Images
Orchard Health Analysis
Apple Detection
Yield Estimation
View Orchard Health Map
View Reports
Download Reports
Login / Logout
Manage Users
Manage Image Dataset
System Configuration
View All Reports
Export Data
Farmer
Register/Login
Upload Images
Preprocess Images
Health Analysis
Apple Detection
Yield Estimation
Health Map
View Reports
Download Reports
Admin
Login/Logout
Manage Users
Manage Dataset
System Config
View All Reports
Export Data

3. Technical Execution (DevOps & Git)

A. **Jira Tasks Completed:**

1. Identify system actors
2. Define system use cases
3. Create UML use case diagram
4. Document use case descriptions

Link:https://ayushshrivastava.atlassian.net/jira/software/projects/DBISFAOM/boards/2?atlOrigin=eyJpIjoiY2I0Y2FjMmQ2MTliNDBlM2FjYzEyN2U1MjdkY2QzNGMiLCJwIjoiaiJ9

B. **Git Activity:**

1. Active Branches: main
2. Merged Pull Requests: Initial documentation setup

C. **CI/CD Status:** Not applicable (design phase – no code yet)

4. Testing Summary

A. **Total Unit Tests:** Not applicable in design phase

B. **Pass Rate:** Not applicable in design phase

C. **Key Bug Identified:** No major issues were identified during this phase.

5. Goals for Next Week

1. Design Class Diagram with attributes and methods
2. Create Object Diagram
3. Start Domain Model / Analysis Classes

Mentor Signature **Project Guide Signature**